

- Q.25 Describe rubber and its properties.
- Q.26 Distinguish between tensile strength & compressive strength.
- Q.27 Write on glass transition temperature.
- Q.28 Describe dielectric strength & dielectric constant.
- Q.29 Explain power loss and how it can be minimized.
- Q.30 Write on specification of material according to BIS.
- Q.31 Write on important varieties of iron ores.
- Q.32 Write the properties and uses of wrought iron.
- Q.33 Explain any two alloys of copper.
- Q.34 Write the properties and use of aluminum.
- Q.35 Describe grey cast iron with their properties and uses.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the importance of engineering materials and their uses.
- Q.37 Classify the iron and explain its types with their properties and uses.
- Q.38 Define phase diagram and describe Fe - C phase diagram.

No. of Printed Pages : 4

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3rd Sem / Chem

Subject:- Introduction of Engineering Materials/ Engg. Materials

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is a non-metal.
- a) Iodine b) Graphite
- c) Carbon d) All of the above
- Q.2 Corrosion of Metals depends upon
- a) Nature of metals
- b) Environment around the metals
- c) Both a and b
- d) None of the above
- Q.3 The Mass per unit volume of a material is called.
- a) Density b) Specific volume
- c) Specific gravity d) None of the above
- Q.4 The material in which atoms are arranged chaotically is called
- a) Crystalline Material b) Amorphous Material
- c) Mesomorphs d) None of the above

(60)

(4) 180532/120532/030523

/531

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/531

Q.5 The ratio of change in dimension of the body to the original dimension of the body is called.

- a) Load b) Silver
- c) Strain d) Fatigue

Q.6 Which of the following is an alloy?

- a) Gold b) Silver
- c) Aluminum d) Steel

Q.7 Brasses are the alloys of

- a) Copper-tin b) Copper-nickel
- c) Copper-nickel - zinc d) Copper-Zinc

Q.8 The purest form of iron is

- a) Pig iron b) cast iron
- c) Steel d) Wrought iron

Q.9 While hardening of steel, the component is cooled in

- a) Still air b) Water or oil
- c) Furnace d) None of the above

Q.10 A Steel with carbon _____ is called with hypoeutectoid steel

- a) 0.8% b) above 0.8%
- c) below 0.8% d) None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 The polymer which can be softened on heating and hardened on cooling is called.

Q.12 Write the Full Form of P.V.C

Q.13 The important use of Bakelite is in the manufacturing of _____

Q.14 Define rubber.

Q.15 Define refractory materials.

Q.16 Write the composition of glass.

Q.17 Name different types of ceramics.

Q.18 Porcelain is a/an _____ of electricity. (Conductor /Insulator)

Q.19 Vanadium when added to steel it _____ tensile strength (Increase/Decrease)

Q.20 Silicon steel is widely use in _____ Industries.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the manufacturing of Nylon 6 and their uses.

Q.22 Write the uses and properties of polypropylene.

Q.23 Define ceramics and write the properties of clay.

Q.24 Define glass and describe Borosilicate glass.